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Subject : Roll No. 26

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Experiment - 1

Objective - Write an assembly language program to add two 8 bit numbers and to subtract two 8 bit number.

Apparatus Required - 8085 microprocessor kit.

To add two 8 bit numbers -

Algorithm -

1. Move first data into accumulator.
2. Move second data in a general purpose register.
3. Add the content of accumulator to the content of general purpose register.
4. Store the result into memory.
5. Stop

Program -

Label	Memory Address	Mnemonics	Hexcode
1	3000H	MVI A 05H	3E
2	3001 H	—	05
3	3002 H	MVI B 04H	06
4	3003 H	—	04
5	3004 H	ADD B	80
6	3005 H	STA 3500H	32
7	3006 H	—	00
8	3007 H	—	35
9	3008 H	HLT	76

Teacher's Signature : _____

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No. 2

Input: 05, 04
Output: 09 stored at location 2050

To subtract two 8 bit numbers-

Algorithm - 1. Move First data into accumulator
2 Move second data in a good general purpose register.
3 Subtract the content of accumulator to the content of general purpose register
4 Store the result into memory
5. Stop

program

Memory Address	Mnemonics	Hex code
3200H	MVI A, 0A	3E
3201H	-	0A
3202H	MVI B 12	06
3203H	-	12
3204H	SUB B	90
3205H	STA 3750 H	32
3206H	-	50
3207H	-	37
3208H	HLT	76

Input: 0A , 12
output F8, Stored at location 3750H

Teacher's Signature :

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Experiment No..... Experiment Result.....

Experiment No - 2

Objective- Write an assembly language program to add two 8 bit numbers stored at two different address.

Apparatus Required- 8085 microprocessor kit.

Algorithm- Let there are two values at the address 3500H & 3520H

20 3500 = 45

3520 = 12

1. Load the content of first address into accumulator.
2. Move the content of accumulator in a general purpose register.
3. Again load the content of second address into accumulator
4. Add the content of accumulator to the content of general purpose register.
5. Store the result into memory
6. Stop the program execution.

Teacher's Signature : _____

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Memory Address	Mnemonics	Hex Code
3100 H	LDA 3500H	3A
3101 H	—	00
3102 H	—	35
3103 H	MOV C, A	4F
3104 H	LDA 3520H	3A
3105 H	—	20
3106 H	—	35
3107 H	ADD C	81
3108 H	STA 3610H	32
3109 H	—	10
310A H	—	36
310B H	HLT	76

Result

Input Address	Value
3500	45
3520	12

Output Address	Value
3610	57

Conclusion - The assembly language program for addition of two number was executed successfully by using 8085 microprocessor kit.

Teacher's Signature : _____

Experiment No-3

Objective- Write an assembly language program to subtract two 8-bit numbers located at two different address.

Apparatus Required- 8085 Microprocessor kit.

Algorithm- Let there are two values at address
3600 H = 52
3700 H = 40

Step 1: Load the content of first address into accumulator.

Step 2: Move the content of accumulator in a general purpose register.

Step 3: Again load the content of second address into accumulator.

Step 4: Subtract the content of accumulator to the content of general purpose register.

Step 5: Store the result into memory.

Step 6: Stop the program execution.

Teacher's Signature : _____

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No 7

Program-

Memory address	Mnemonics	Hex code
2100 H	LDA 3600H	3A
2101 H	-	00
2102 H	-	36
2103 H	MOV C,A	4F
2104 H	LDA 3700H	3A
2105 H	-	00
2106 H	-	37
2107 H	SUB C	91
2108 H	STA 3020H	32
2109 H	-	20
210A H	-	30
210B H	HLT	76

Result-

Input Address	Value
3600	52
3700	40

Output Address	Value
3020	12

Conclusion - The assembly language program for subtraction of two numbers was executed successfully by using 8085 Microprocessor kit.

Teacher's Signature : _____

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No. 8

Experiment No - 4

Objective- To exchange the contents of two memory locations

Apparatus Required - 8085 Microprocessor kit.

Algorithm-

Step 1: Load the accumulator with the content of one location.

Step 2: Move the content of accumulator to another register.

Step 3: Load the accumulator with the content of second register

Step 4: Store the content of accumulator at first memory location.

Step 5: Move the content of previous register to accumulator

Step 6: Store the content of accumulator at second memory location.

Step 7: Stop the program execution.

To exchange. 2500H = 15 & 2501 H = 13

Teacher's Signature : _____

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Program :

Memory Address	Mnemonics	Hex Code
2000 H	LDA 2500H	3A
2001 H	-	00
2002 H	-	25
2003 H	MOV C,A	4F
2004 H	LDA 2501H	3A
2005 H	-	01
2006 H	-	25
2007 H	STA 2500H	32
2008 H	-	00
2009 H	-	25
200A H	MOV A,C	79
200B H	STA 2501H	32
200C H	-	01
200D H	-	25
200E H	HLT	76

Result

Input Address	Value
2500H	15
2501H	13
Output Address	**Value**
2500H	13
2501H	15

Conclusion- The assembly language program for exchange the contents of two memory locations executed successfully

Teacher's Signature : _____

Experiment - 5

Objective - Write an assembly language program to add two 16 bit numbers and to subtract two 16 bit numbers.

Algorithm - 1. Move a 16 bit number from memory to HL pair register.
2. Move another 16-bit number from memory to general purpose register pair (BC or DE)
3. Add the content of HL pair to the content of general purpose register pair.
4. Store the result in memory from HL pair
5. Stop

Program

Memory Address	Mnemonics	Hex Code
3100H	LXI H 3050H	21
3101 H	-	50
3102 H	-	30
3103 H	LXI B 1234 H	01
3104 H	-	34
3105 H	-	12
3106 H	DAD B	09
3107 H	SHLD 3060	22
3108 H	-	60
3109 H	-	30
310A H	HLT	76

Input: 3050 & 1234
Output: 4284

Teacher's Signature :

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No. 11

2

Objective - Write an assembly language program to add content of two 16 bit numbers.

Algorithm - 1. Load register pair HL from memory location 3050.
2. Exchange the content of HL pair register with DE register.
3. Again load register pair HL from memory location 3055
4. Add the content of DE registers with HL pair register.
5. Store the result in memory from HL pair register
6. Stop.

Program.

Memory Address	Mnemonics	Hex code
3200H	LHLD 3050	2A
3201H	—	50
3202H	—	30
3203H	XCHG	EB
3204H	LHLD 3050	2A
3205H	—	55
3206H	—	30
3207H	DAD D	19
3208H	SHLD 3070	22
3209H	—	70
320AH	—	30
320BH	HLT	76

Input 3050 [2A] 3051 [43]
3055 [12] 3056 [3C]

Output. 3070 [3C]
3071 [7F]

Teacher's Signature : _____

Experiment No – 8

Objective – Write an assembly language program to add two 8 bit numbers using LXI command.

Algorithm –

Step 1: Load the content from memory (using two 8 bit numbers) into HL pair register move

Step 2: Move the content of HL pairs into accumulator.

Step 3: Move the content of accumulator in a general purpose register.

Step 4: Increment the content of H by 1

Step 5: Move the content of HL pair register into accumulator
Add the content of accumulator

Step 6: Add the content of accumulator to the of general purpose register.

Step 7: Increment the content of H by 1.

Step 8: Move the content of accumulator to register H

Step 9: Stop

Teacher's Signature : _____

Name of Experiment..... Date.....

Experiment No..... Experiment Result.....

Program

Memory address	Mnemonics	Hexcode
3300 H	LXI H 3080	21
3301 H	—	80
3302 H	—	30
3303 H	MOV A, M	7E
3304 H	MOV B, A	47
3305 H	INX H	23
3306 H	MOV A, M	7E
3307 H	ADD B	80
3308 H	INX H	23
3309 H	MOV M, A	77
330A H	HLT	76

Input 3080:2A
3081:43

Output 3082:6D

Teacher's Signature :

Memory Address	Mnemonics	Hex code
4000H	LDA 4400H	3A
4001H	—	00
4002H	—	44
4003H	CMA	2F
4004H	STA 4300H	32
4005H	—	00
4006H	—	43
4007H	HLT	76

Experiment-7

Object- Find the 1's complement of the number stored at memory location 4400H and store the complemented number at memory location 4300H.

Apparatus Required- 8085 microprocessor kit.

Algorithm-

1. Load the data from memory 4400H into accumulator
2. Complement content of accumulator.
3. Store the content of accumulator in memory 4300H (1's complement)
4. Stop

Program

Label	Memory Address	Mnemonics	Comment
1	4000H	LDA 4400H	[A] ← [4400]
2	4003H	CMA	[A] ← [A]
3	4004H	STA 4300H	1's complement
4	4007H	HLT	stop

Input- 05
Output- FA

Teacher's Signature :

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No. 15

Experiment NO-8

Objective - Find the 2's complement of the number stored at memory location 4200H and store the complemented number at memory location 4300H.

Apparatus Required - 8085 microprocessor kit

Algorithm.

1. Load the data from memory 4200H into accumulator.
2. Complement content of accumulator.
3. Add 01 to Accumulator content.
4. Store content of accumulator in memory 4300H (2's complement)
5. Stop

Program

Label	Memory Address	Mnemonics	Hex code
1	4100 H	LDA 4200H	3A
2	4101 H		00
3	4102 H		42
4	4103 H	CMA	2F
5	4104 H	ADI, 01	C6
6	4105 H		01
7	4106 H	STA 4300H	32
8	4107 H		00
9	4108 H		43
10	4109 H	HLT	76

Input - 05

Output - FB

Teacher's Signature :

Memory Address	Mnemonics	Hex code
2000	LDA 4201H	3A
2001		01
2002		42
2003	RLC	07
2004	RLC	07
2005	RLC	07
2006	RLC	07
2007	MOV C, A	4F
2008	LDA 4200	3A
2009		00
200A		42
200B	ADD C	89
200C	STA 4300H	32
200D		00
200E		43
200F	HLT	76

Experiment No-9

Object- Pack the two unpacked BCD numbers stored in memory locations 4200H and 4201 and store result in memory location 4300H. Assume the least significant digit is stored at 4200H

Apparatus Required- 8085 microprocessor kit.

Program.

```
LDA  4201H   : Get the most significant bit
RLC
RLC
RLC
RLC          : Adjust the position of second digit
MOV C,A      : Store the partial result
LDA 4200H    : Get the lower BCD digit
ADD C        : Add lower BCD digit
STA 4300H    : Store the result
HLT          : Terminate program execution
```

Input- 05
04

Output- 54

Teacher's Signature :

Memory Address	Mnemonics	Hex Code
4000	LDA 4200	3A
4001		00
4002		42
4003	ANI F0H	E6
4004		F0
4005	RRC	0F
4006	RRC	0F
4007	RRC	0F
4008	RRC	0F
4009	STA 4301	32
400A		01
400B		43
400C	LDA 4200	3A
400D		00
400E		42
400F	ANI 0F	E6
4010		0F
4011	STA 4300	32
4012		00
4013		43
4014	HLT	76

Experiment No - 10

Object - Two digit BCD number is stored in memory location 4200H Unpack the BCD number and store the two digits in memory locations 4300H and 4301H such that memory location 4300H will have lower BCD digit.

Apparatus Required - 8085 Microprocessor Kit

Program -

```
LDA   4200H ; Get the packed BCD number.
ANI   F0H ; Mask lower order bits (4 bits)
RRC
RRC
RRC
RRC                ; Adjust higher BCD digit as a lower digit
STA   4301H        ; Store the partial result
LDA   4200H        ; Get the original BCD number
ANI   0FH          ; Most higher order bits (4 bits)
STA   4300H        ; Store the result
HLT                ; Terminate program execution
```

Input - 0054

Output - 0504
01

Teacher's Signature :

Name of Experiment..... Date.....

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Experiment No - 11

Object - Write a set of instructions to alter the contents of flag register in 8085.

Apparatus Required - 8085 microprocessor kit

Program -

Address	Mnemonics		Hex Code
2000	PUSH PSW	; Save flags on stack	F5
2001	POP H	; Retrieve flags in L	E1
2002	MOV A, L	; Flags in accumulator	7D
2003	CMA	; Complement accumulator	2F
2004	MOV L, A	; Accumulator in 'L'	6F
2005	PUSH H	; Save on stack	E5
2006	POP PSW	; Back to flag register	F1
2007	HLT	; Terminate program execution	76
2008			

Result :-

Teacher's Signature :

Address	Mnemonics	Hex Code
4000	LDA 4200	3A
4001	-	00
4002	-	42
4003	MOV C,A	4F
4004	SUB A	97
4005	LXI 4201	21
4006		01
4007		42
4008	ADD M	86
4009	INX H	24
400A	DCR C	0D
400B	JNZ 4009	C2
400C		09
400D		40
400E	STA 4300	32
400F		00
4010		43
4011	HLT	76

Experiment No- 12

Object - Calculate the sum of series of numbers. The length of the series in memory location 4200H and the series begins from memory location 4201H consider the sum to be 8 bit number ignore carries store the sum at memory location 4300H.

Apparatus Required - 8085 microprocessor kit.

Program -

```
LDA 4200H
MOV C,A     ; Initialize counter
SUB A       ; Sum = 0
LXI H, 4201H ; Initialize pointer
abc: ADD M  ; Sum = sum + data
   INX H    ; Increment pointer
   DCR C    ; Decrement counter
   JNZ abc  ; if counter not 0 repeat
   STA 4300H ; Store Sum
   HLT      ; Terminate program execution.
```

Resul- The program is executed successfully

Teacher's Signature : _____

Label	Memory Address	Mnemonics	Hexcode
1	2001H	LDA 2500H	3A
2	2002H	—	00
3	2003H	—	25
4	2004H	MOV B, A	47
5	2005H	MOV C, A	4F
6	2006H	XRA A	AF
7	2007H	ADD B	80
8	2008H	DCR C	0D
9	2009H	JNZ 2007H	C2
10	200AH	—	07
11	200BH	—	20
12	200CH	STA 2600	32
13	200DH	—	00
14	200EH	—	26
15	200FH	HLT	76

Name of Experiment..... Date.....

Experiment No..... Experiment Result..... Page No. 20

Experiment No-13

Object:- Write a program to calculate square of a given number.

Apparatus Required:- 8085 Microprocessor kit

Program

```
LDA 2500H     Assuming some 8-bit data at 2500H
MOV B,A       Load data to accumulator
              move data from accumulator to B
              register
MOV C,A       Loop
XRA A         Make content of accumulator 00
LOOP: ADD B   Add content of accumulator with
              B register
DCR C         Decrement C register
JNZ LOOP      Checking for no zero
STA 2600H     Store result at 2600H memory
              location
HLT           Stop
```

Result:

Input: 2500H 05H

Output: 2600H 19H

Teacher's Signature : _____

Label	Memory Address	Mnemonics	Hex Code
1	2000H	LDA 2050H	3A
2	2001H	—	50
3	2002H	—	20
4	2003H	MOV B, A	47
5	2004H	LDA 2051H	3A
6	2005H	—	51
7	2006H	—	20
8	2007H	MOV C, A	4F
9	2008H	XRA A	AF
10	2009H	ADD B	80
11	200AH	DCR C	0D
12	200BH	JNZ 2009H	C2
13	200CH	—	09
14	200DH	—	20
15	200EH	STA 2056	32
16	200FH	—	56
17	2010H	—	20
18	2011H	HLT	32

Experiment No - 14

Objective: Write a program to multiply two numbers.

8085 microprocessor kit.

Program: Assuming first 8 bit number at 2050H & second 8 bit number at 209H

```
LDA 2050H    Load first number into accumulator
MOV B,A      Move content of accumulator to B register
LDA 209H     Load second number into accumulator
MOV C,A      move content of accumulator to C register
XRA A        make content of accumulator 0
xy3: ADD B   Add content of B register
DCR C        Decrement C register
JNZ xy3      Condition
STA 2056H    Store result at 2056H memory location
HLT          Stop
```

Result

Input - 2050 H [7H]
209 H [3H]

Output 2056 H [15H]

Teacher's Signature : _____

Memory Address	Mnemonics	Hexcode
2000	LDA 2100	3A
2001	—	00
2002	—	21
2003	MVI B,00	06
2004	~~MVI C~~ —	00
2005	MVI C,08	06
2006	—	08
2007	RAL	17
2008	JNC 2000	D1
2009	—	00
200A	—	20
200B	INR B	04
200C	DCR C	0D
200D	JNZ 2007	C2
200E		07,20
2010	MOV A,B	~~200~~ 78
2011	STA 2200	32
2012		00
2013		22
2014	HLT	76

Experiment No-15

Objective:- Write a program to count no of ones in a data byte.

Flow chart (or) algorithm:-

1. Load data in accumulator from memory
2. Move/initialize register B with data '00'
3. Initialize register C with data '00'
4. Rotate content of A with carry to left
5. Jump if no carry to decrement operation
6. Increment contents of register B
7. Decrement contents of register C
8. Jump if not zero to rotate the data of A
9. Move contents of B in A
10. Store the contents at location 2200
11. Stop

Input:- Number at address 2100

Output:- The no of one's in the content of 2100 is stored at address 2200

Teacher's Signature :

Memory Address	Mnemonics	Hexcode
3000	XRAA	AF
3001	LXIH 2050	21
3002	—	50
3003	—	20
3004	MVI C,06	0E
3005	—	06
3006	MOV B,A	47
3007	ADDM	86
3008	INC	02
3009	—	0C
300A	—	30
300B	INXH	23
300C	DCRC	0D
300D	JNZ	C2
300E	—	07
300F	—	30
3010	STA3100	32
3011	—	00
3012	—	31
3013	HLT	76

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Experiment No..... Experiment Result.....

Experiment No - 16

Objective: 6 bytes of data are stored in memory starting at 2050H. Add all the data bytes. Use register B to store any carry generated

Apparatus Required- 8085 microprocessor kit

Program

```
XRAA          Make content of accumulator 0
LXIH 2050     Load the content into HL pair
MVIC,06
MOV B,A       move data from accumulator to B register
ADD M         Add content
JNC           Checking for no zero carry
INXH          Increment pointer
DCR C         Decrement counter
JNZ           Checking for no zero
STA 3100H     store result at 3100H memory location
HLT           Stop
```

Teacher's Signature : _____

Memory Address	Mnemonics	Hexcode
3000	MVI B, 00	06
3001		00
3002	MVI C, 05	0E
3003		05
3004	LXI H 2000	21
3005	—	00
3006	—	20
3007	MOV A, M	7E
3008	ANI 0, 1	E6
3009	—	01
300A	JNZ 300B	C2
300B	—	0E
300C	—	30
300D	INR B	04
300E	INX H	23
300F	DCR C	0D
3010	JNZ 3007	C2
3011	—	07
3012	—	30
3013	MOV A, B	78
3014	STA 3500	32
3015	—	00
3016	—	35
3017	HLT	76

Experiment No. 17

Objective:- Write an assembly language program to count even numbers from a series.

Flow Chart or Algorithm:-

1. Store five numbers from location 2000 to 2004
2. Start the counter using register C as 5
3. Move the number from memory to accumulator one by one
4. Perform ANI operations between the content of accumulator and 01
   If answer is zero increase the counter B by 1 everytime we check from series.
5. Store the final result of B at any memory location.
6. Stop the program

Input:- Series of 5 numbers

Output: total count of even no in a series is stored at 3500

Teacher's Signature : _____

Memory address	Mnemonics	Hexcode
2000	LXIH 2500	21
2001	—	00
2002	—	25
2003	MVIC 00	0E
2004	—	00
2005	MVID, 0A	16
2006	—	0A
2007	INXH	23
2008	MOV A, M	7E
2009	ANI, 01	E6
200A	—	01
200B	JZ 200F	CA
200C	—	0F
200D	—	20
200E	INR C	0C
200F	DCR D	15
2010	JNZ 2007	C2
2011	—	07
2012	—	20
2013	MOV A, C	79
2014	STA 3050	32
2015	—	50
2016	—	30
2017	HLT	

Objective:- Write a program to count odd numbers from a series.

Flow chart (or) algorithm:-

1. Load the content (data) from memory into HL register pair.
2. Move 0 in register C (counter)
3. Move 0A in register D (size of series)
4. Increment the content of HL register pair.
5. Move the content of HL register pair to accumulator AND it with 1
6. If the result is not zero decrement the counter of register D
7. If the result is zero repeat the process after incrementing C and decrementing the content is 0.
8. If the content of D is not zero move the value in C to accumulator.
9. Store the result in memory
10. & stop

Input:- Content at memory location 04, 03, 00, 09, 01, 02
05, 06, 00

Output: Stored at location 3050 :- 5

Teacher's Signature : _____